Pranjal Singh

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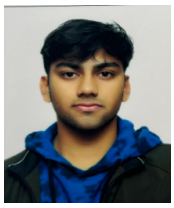

Education

Manipal Institute of Technology — Manipal, India
*B.Tech in Computer Science and Engineering – **CGPA: 9.25/10*** — *Aug 2024 – May 2028*
Relevant Coursework: DSA · Machine Learning · Database Systems · Computer Vision · Operating Systems · AI · OOP

ASN International School — India
Class XII (CBSE) – 95.6% — *2024*

Father Agnel School, Noida — India
Class X (CBSE) – 99% — *2022*

Projects

Stock Heroes | *React, TypeScript, REST APIs, Yahoo Finance, localStorage* — 2025
- Built a financial simulation platform with virtual NSE stock trading, real-time price integration via Yahoo Finance API, intraday charts, and portfolio P&L tracking.
- Engineered a custom `useStockData` hook with 60s polling, offline fallback, and full `localStorage` state persistence across browser sessions.
- Designed a gamified quiz engine awarding virtual cash rewards, a dynamic leaderboard with live rank updates, and LLM-powered financial literacy challenges.
- Implemented intelligent routing and agent selection logic; end-to-end ownership from architecture to deployment on Vercel.

PandavAI | *React, LLM APIs, Prompt Engineering, Multi-Agent Systems* — 2025
- Developed a multi-agent conversational AI system where each agent specialises in a distinct domain: knowledge, creativity, skills, and life guidance.
- Implemented intelligent query routing to direct user inputs to the most contextually appropriate agent; integrated LLM APIs with advanced prompt engineering for contextual responses.
- Designed an interactive React interface supporting dynamic multi-turn conversations and real-time agent switching.

Real-Time Vehicle Speed Detection | *Python, OpenCV, Computer Vision* — 2025
- Built a CV pipeline estimating vehicle speeds from live camera feeds using background modeling, frame differencing, grayscale processing, and morphological filtering.
- Designed zone-based vehicle tracking algorithms to calculate speed across detection boundaries; built an interactive dashboard showing live stats and violation logs.

Technical Skills

Languages: Java, Python, C, C++, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks & Libraries: React, Node.js, Flask, Flutter, Tailwind CSS, Recharts, Radix UI

AI / ML: TensorFlow, scikit-learn, OpenCV, LLM API Integration, Prompt Engineering, Multi-Agent Systems

Data & Tools: Pandas, NumPy, Matplotlib, Seaborn, Git, Vite, VS Code, Jupyter, REST APIs, Real-Time Data Pipelines

Concepts: DSA, OOP, Computer Vision, Machine Learning, Agile Development, CI/CD Basics, Client-Side State Management

Extracurricular

HFS Organising Committee | *Revels & Techtatva, MIT Manipal* — 2024 – Present
- Part of the Hospitality & Food Services organising committee for MIT's flagship cultural fest (Revels) and technical fest (Techtatva), coordinating logistics for 10,000+ attendees.

Swimming Team | *MIT Manipal* — 2024 – Present
- Competitive swimmer representing MIT Manipal at intercollegiate aquatics events.

Achievements

- **National-level swimmer** – represented at national aquatics competitions.
- **State-level basketball player** – selected for state-level representation.
- **National Karate medalist** – medal winner at national karate championships.
- **Multiple Academic Excellence Awards** – consistent top performer; 99% in Class X, 95.6% in Class XII, 9.25 CGPA at MIT Manipal.